

## 1 What is Gradient Descent

When we break the word Gradient Descent and understand what it actually means, gradient means inclined part of a pathway or a slope, descent means to move towards the bottom of the slope. Imagine yourself to be a mountaineer, and you are trying to get to the bottom of the mountain, you are descending the gradient of a mountain.

## 2 Understanding Gradient Descent for $f(x) = x^2$

Suppose function  $y = f(x)$ , where  $x, y$  are real numbers.

- This function has minimum at  $x = 0$  which we want to determine using gradient descent.
- Derivative of function denoted:  $f'(x)$  or as  $\frac{dy}{dx}$
- Derivative  $f'(x)$  gives the slope of  $f(x)$  at point  $x$ .
- It specifies how to scale a small change in input to obtain a corresponding change in the output:  
 $f(x + \eta) \approx f(x) + \eta f'(x)$  where  $\eta$  is a small change made.
- It tells how you make a small change in input to make a small improvement in  $y$ .
- We know that  $f(x - \eta \text{sign}(f'(x)))$  is less than  $f(x)$  for small  $\eta$ . Thus we can reduce  $f(x)$  by moving  $x$  in small steps with opposite sign of derivative

$$\text{where } \text{sign}(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}$$

This technique is called gradient descent.

Consider a simple math equation  $f(x) = x^2$ ,

| x  | $f(x) = x^2$ |
|----|--------------|
| 1  | 1            |
| 2  | 4            |
| 3  | 9            |
| 4  | 16           |
| 5  | 25           |
| 6  | 36           |
| 0  | 0            |
| -1 | 1            |
| -2 | 4            |
| -3 | 9            |
| -4 | 16           |
| -5 | 25           |
| -6 | 36           |

When above values are plotted we get the following graph

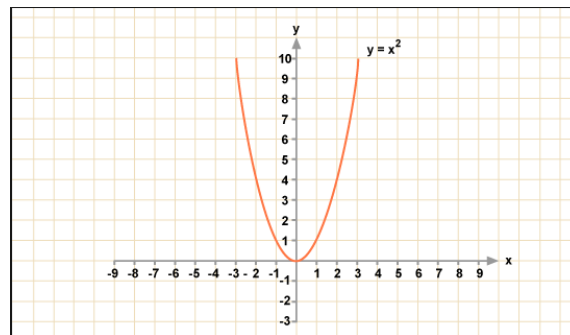


Figure 1

Here, using gradient descent we want to find the minimum of the function  $f(x)$ .

This function has minimum at  $x = 0$  which we want to determine using gradient descent.

We have  $f'(x) = 2x$

For gradient descent, we update by  $-f'(x)$

If  $x(t) > 0$  then  $x(t+1) < x(t)$

If  $x(t) < 0$  then  $f'(x) = 2x$  is negative, thus  $x(t+1) > x(t)$

**Update rule:**  $x' \leftarrow x - \eta \frac{dy}{dx}$

#### Procedure:

```

Gradient-Descent(
   $x'$  //Initial starting point
   $f$  //function to be minimized
   $\delta$  //Convergence threshold )
1  $t \leftarrow 1$ 
2 do
3  $x(t+1) \leftarrow x(t) - \eta \frac{dy}{dx}$ 
4  $t \leftarrow t+1$ 
5 while  $\|x(t+1) - x(t)\| > \delta$ 
6 return ( $x(t)$ )
  
```

## References:

Gradient descent over multi-dimensional parameters:

<https://mxnet.apache.org/versions/1.9.1/api/python/docs/tutorials/packages/optimizer/index.html>

An Overview of Gradient Descent Algorithm:

<https://www.ruder.io/optimizing-gradient-descent/>